



The Salinity of the Ocean — FULL MIND MAP

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Chapter 23_The_Salinity_of_the_Ocean

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Salinity Gradient Terminology

- **Halocline** = water salinity gradient (depth ke saath salinity rapidly change) — Thermocline (temperature change), Pycnocline (density change) se contrast. [Q1]

Salinity ka Main Source

- Sea salinity ka main source **Land** hai (rivers various salts sea tak le jaati hain) — other factors: vaporization, wind, rain, ocean currents, volcanoes. Average ocean salinity ~35‰. [Q2]
- Salinity mein sabse zyada contribution **Sodium Chloride** (77.8%) ka hota hai — William Dittmar's 1884 Challenger expedition report (47 types of salts, 7 prominent). [Q3]

Highest Salinity Water Bodies

- **Dead Sea** = world ki sabse zyada saline water body examples mein (~34%/340‰) — Lake Van Turkey (~22‰), Great Salt Lake (variable), Caspian Sea, Black Sea, Baltic Sea, Yellow Sea, Mediterranean sab isse KAM saline hain. [Q4, Q5, Q6, Q7, Q8, Q12]

Density-Salinity-Depth Relationship (Exam Convention)

- Accepted exam relation: sea mein density badhne par **salinity badhti hai** lekin **depth ghat-ti hai**. [Q9]

Great Salt Lake

- **Great Salt Lake** = USA (Utah state) mein hai — salinity highly variable hoti hai (lake level aur arm ke hisaab se), fixed value se represent nahi kiya jaana chahiye. [Q10]

Arabian Sea Salinity

- Arabian Sea ki average salinity ~**35 ppt (36‰)** hai — dry weather + high evaporation + kam river inflow ki wajah se; Bay of Bengal isse kam (~30‰) hai. *[Q11]*